

Trabajo Practico Estructura de Datos

2) a) 1)

sum=0; ← 1OE

for(i=1; i<=n; i++) ← 4OE

sum=sum + 1; ← 2OE

$$T(n) = 1 + 1 + 1 + \sum_{i=1}^n (2 + 1 + 2)$$

$$T(n) = 3 + \sum_{i=1}^n 5$$

$$T(n) = 3 + (n - 1 + 1) * 5$$

$$T(n) = 3 + (n + 0) * 5$$

$$T(n) = 3 + 5n \text{ - COMPLEJIDAD LINEAL}$$

2)

sum=0; ← 1OE

for(i=1; i<=n; i++) ← 4OE

for(j=1; j<=n; j++) ← 4OE

sum=sum + 1; ← 2OE

$$T(n) = 1 + 1 + 1 + \left[\sum_{i=1}^n 1 + 1 + \left(\sum_{j=1}^n 2 + 1 + 2 \right) + 1 + 2 \right]$$

$$T(n) = 3 + \left[\sum_{i=1}^n 2 + \left(\sum_{j=1}^n 5 \right) + 3 \right]$$

$$T(n) = 3 + \left[\sum_{i=1}^n 2 + ((n - 1 + 1) * 5) + 3 \right]$$

$$T(n) = 3 + \left[\sum_{i=1}^n 2 + ((n - 0) * 5) + 3 \right]$$

$$T(n) = 3 + \left[\sum_{i=1}^n 2 + 5n + 3 \right]$$

$$T(n) = 3 + \left[\sum_{i=1}^n 5 + 5n \right]$$

$$T(n) = 3 + (n - 1 + 1) * (5 + 5n)$$

$$T(n) = 3 + (n - 0) * (5 + 5n)$$

$$T(n) = 3 + 5n^2 + 5n$$

$$T(n) = 5n^2 + 5n + 3 \text{ - COMPLEJIDAD CUADRATICA}$$

3)

sum=0; ← 1OE

for(i=1; i<=n; i++) ← 4OE

for(j=1; j<=n * 2; j++) ← 5OE

sum=sum + 1; ← 2OE

$$T(n) = 1 + 1 + 1 + \left[\sum_{i=1}^n 1 + 2 + \left(\sum_{j=1}^{n*2} 2 + 2 + 2 \right) + 1 + 2 \right]$$

$$T(n) = 3 + \left[\sum_{i=1}^n 3 + \left(\sum_{j=1}^{n*2} 6 \right) + 3 \right]$$

$$T(n) = 3 + \left[\sum_{i=1}^n 3 + ((n * 2 - 1 + 1) * 6) + 3 \right]$$

$$T(n) = 3 + \left[\sum_{i=1}^n 3 + (2n * 6) + 3 \right]$$

$$T(n) = 3 + \left[\sum_{i=1}^n 3 + 12n + 3 \right]$$

$$T(n) = 3 + \left[\sum_{i=1}^n 3 + 12n + 3 \right]$$

$$T(n) = 3 + \left[\sum_{i=1}^n 12n + 6 \right]$$

$$T(n) = 3 + [(n - 1 + 1) * (12n + 6)]$$

$$T(n) = 3 + [(n + 0) * (12n + 6)]$$

$$T(n) = 3 + 12n^2 + 6n$$

$$T(n) = 12n^2 + 6n + 3 - \text{COMPLEJIDAD CUADRATICA}$$

b)

	Tiempo ejecución 1000(nano)	Tiempo ejecución con 100000 (nano)
Punto 1	1297414096153400	1294923488176900
Punto 2	1297473952925300	1297498528426500
Punto 3	1297570448629800	1297588534956200

4)

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#include <stdio.h>
#include <stdlib.h>
int main(int argc, char **argv) {
    int arreglo[100], i, N, pos=0, mayor; ← 1OE
    printf("Ingrese el valor de N: "); ← 1OE
    scanf("%d", &N); ← 2OE
    if (N < 1 || N > 100){ ← 3OE
        printf("Error, tiene que ser entre 1 y 100."); ← 1OE
        return 0; ← 1OE
    }
    else {
        for (i=0; i < N; i++) { ← 4OE
            arreglo[i]= rand()% 10; ← 4OE
            printf("%d", arreglo[i]); ← 3OE
        }
        mayor = arreglo[pos]; ← 2OE
        for (i=1; i < N; i++) ← 4OE
            if(arreglo[i] > mayor){ ← 2OE
                pos = i; ← 1OE
                mayor = arreglo[pos]; ← 2OE
            }
        printf("\nEl %d, en la posición: %d", mayor, pos); ← 3OE
    }
}
return 0; ← 1OE
}

```

$$T(n) = 1 + 1 + 2 + 3 + [1 + 1 + \sum_{i=0}^{n-1} 4 + 3 + 1 + 2] + 2 + [1 + 1 + \sum_{i=1}^{n-1} (2 + 1 + 2 + 1 + 2)] + 3 + 1$$

$$T(n) = 7 + (2 + \sum_{i=0}^{n-1} 10) + 2 + (2 + \sum_{i=1}^{n-1} 8) + 4$$

$$T(n) = 7 + [2 + (n - 1 - 0 + 1) * 10] + 2 + [2 + (n - 1 - 1 + 1) * 8] + 4$$

$$T(n) = 7 + (2 + n * 10) + 2 + [2 + (n - 1) * 8] + 4$$

$$T(n) = 7 + 2 + 10n + 2 + 2 + 8n - 8 + 4$$

$$T(n) = 18n + 9 - \text{COMPLEJIDAD LINEAL}$$